Tahmidul Bin Ferdous

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RESEARCH INTERESTS

Primary: Technical AI Safety, Alignment, Evaluation Validity, Benchmark Integrity & Data Contamination Auditing.
Secondary: Generative Modeling (Flow Matching, Diffusion, GANs), Medical Computer Vision, Distributed Training (Multi-GPU DDP, TPU v5e/JAX).

EDUCATION

International Islamic University Chittagong (IIUC) *2022 – 2026*
Bachelor of Science in Computer Science and Engineering | Chittagong, Bangladesh

- **Cumulative CGPA: 3.311 / 4.00** | **Credits Completed: 163 / 163** (all degree requirements fulfilled).
- **Academic Trajectory & Honors:** Strong upward performance trend, graduating with **GPA 3.702 / 4.00** in the final semester.
- **Undergraduate Thesis Grade:** Awarded highest departmental grade: **Grade A** (4.00 / 4.00, 4.0 credit load).
- **Relevant Coursework:** Neural Networks & Fuzzy Systems (**A**, Lab **A**), Pattern Recognition & Image Processing (**A−**, Lab **A+**), Computer Security (**A−**, Lab **A+**), Artificial Intelligence (**A−**, Lab **A−**), Mathematical Analysis for CS (**A−**), Technical Writing & Presentation (**A+**).

UNDERGRADUATE THESIS

Auditing Data Leakage in Lung Cancer CT Classification: A Controlled Benchmark and Sensitivity Analysis *2025 – 2026*
Supervisor: Md. Khaliluzzaman, Associate Professor, Department of CSE, IIUC | Grade: A | Defended: July 1, 2026

- **Experimental Design:** Formulated a controlled paired audit across four architecture families (CCAL ~666K, EfficientNet B0–B4 4.0M–17.6M, ViT-B/16 85.8M, PLOS-style CNN ~2M) on the IQ-OTH/NCCD benchmark across 163 logged W&B runs, isolating data partitioning as the sole manipulated variable while fixing architectures, preprocessing, augmentation, and hyperparameters.
- **Empirical Findings:** Proved that replacing random splitting with deterministic ordering splits incurs a robust **≈ 7.5 percentage point drop** in test accuracy (CCAL: 95.80% → 88.29%; EfficientNet: 99.25% → 91.71%) and causes minority-class (Benign) F1 score to **collapse by roughly half (80.7 → 40.4)**, demonstrating that published near-perfect benchmark scores stem from slice-level patient contamination.
- **Analytical Proof:** Derived closed-form leakage probability via hypergeometric distribution: under random 80/20 splitting, an individual patient's CT volume leaks across partitions with $P(\text{leak}) \approx 0.84$, with an expected 12.6 of 15 minority-class cases straddling train and test splits.

PUBLICATIONS & MANUSCRIPTS

[1] **Tahmidul Bin Ferdous**, M. Khaliluzzaman. "How Much of IQ-OTH/NCCD Accuracy Is a Split Artefact? A Controlled Ordering Audit Across Four Architectures." *Manuscript in prep for arXiv (cs.CV, eess.IV) and target submission to MICCAI Workshop on Fairness of AI in Medical Imaging (FAIMI)*, 2026.

[2] **Tahmidul Bin Ferdous**. "Deep Learning for Lung Cancer CT Classification: A Systematic Survey of State-Space Models, Efficient Backbones, and Benchmark Integrity (2023–2026)." *Technical Monograph & Literature Survey, Department of CSE, IIUC*, 2026.

RESEARCH & PROFESSIONAL EXPERIENCE

FlyRank AI *July 2026 – December 2026*
Machine Learning Engineering Intern | Remote (USA)

- Engineer production machine learning pipelines and quantitative ranking models for automated search retrieval and domain analytics.
- Benchmark evaluation workflows, profiling inference latency and optimizing feature preprocessing pipelines across distributed services.

Undergraduate Research Assistant (Thesis) *January 2025 – June 2026*
Department of Computer Science and Engineering, IIUC | Chittagong, Bangladesh

- Led benchmark reproducibility audits; successfully replicated 4 out of 5 published EfficientNet baseline accuracies within ±0.5 percentage points.
- Orchestrated 163 end-to-end model training runs on constrained remote environments, managing gradient logging and multi-seed convergence on W&B.

Outlier AI *November 2024 – February 2025*
AI Data Quality & Alignment Specialist | Remote

- Audited and curated large-scale human preference and reasoning datasets utilized for Reinforcement Learning from Human Feedback (RLHF) and alignment.
- Refined multi-turn annotation guidelines, directly mitigating prompt ambiguity and improving inter-annotator reliability across reasoning benchmarks.

Freelance Contract (USA Client) *November 2024*
Computer Vision Engineer | Remote

- Designed and deployed an automated computer vision packet-counting system for inventory management within a 15-day timeline, achieving real-time execution (>30 FPS) under CPU constraints.

SELECTED RESEARCH & TECHNICAL PROJECTS

Conditional Rectified Flow for CT Synthesis on Google Cloud TPU v5e *Feb 2026 – Mar 2026*

PyTorch, JAX/Flax, Google Cloud TPU v5e, Weights & Biases (18 runs, 32.1 GPU-hours)

- Implemented class-conditional Rectified Flow (flow matching) from original mathematical literature: linear interpolant trajectory, velocity-prediction objective, logit-normal timestep sampling, and Euler/midpoint ODE numerical solvers.
- Built multi-GPU DistributedDataParallel (DDP) PyTorch training infrastructure, then **ported the complete architecture to JAX/Flax for 8-device Google TPU v5e**, achieving **576 images/sec** (2.50 s/epoch vs. 15.5 s/epoch on 2× Tesla T4)—a **~6× throughput speedup**.

BanglaPoemGPT — 10M-Parameter Autoregressive Transformer from Scratch *March 2025*

Python, PyTorch, Kaggle P100 (4 runs, 2.0 GPU-hours)

- Engineered a GPT-1 architecture from first mathematical principles without high-level library wrappers: multi-head causal self-attention, learned positional embeddings, custom tokenizer, and autoregressive generation loop.
- Trained on a 3M-token Bengali poetic corpus, achieving validation cross-entropy loss of **1.693** with coherent generative sampling.

Conditional DDPM for Minority-Class CT Balancing *February 2026*

PyTorch, DistributedDataParallel, 2× Tesla T4 (5 runs, 4.2 GPU-hours)

- Constructed a class-conditional Denoising Diffusion Probabilistic Model (DDPM) featuring a conditional U-Net backbone, cosine noise schedule, and multi-GPU DDP orchestration, reaching a training loss of **0.0238** over 3,000 epochs.

Edge Vision: YOLOv11 Campus Security & Facial Biometric Systems *Oct 2024 – Nov 2024*

YOLOv11, OpenCV, FaceNet, PyTorch

- Curated, annotated, and trained custom YOLOv11-s/m models for campus ID compliance monitoring, reaching **mAP50 = 0.82**.
- Implemented edge facial recognition utilizing SSD face detectors and FaceNet 128-d embeddings with cosine similarity matching, calibrating decision thresholds against false-accept/false-reject trade-offs.

TECHNICAL SKILLS

Programming Languages: Python (Primary, 250+ logged experiments), C++, SQL, Bash/Shell, LaTeX.

Deep Learning Frameworks: PyTorch (Advanced), JAX/Flax, torchvision, timm, TensorFlow/Keras, Hugging Face.

Computer Vision & ML Libraries: OpenCV, MediaPipe, Ultralytics YOLOv11, scikit-learn, Albumentations, NumPy, Pandas, Matplotlib.

Systems & Infrastructure: Distributed Data Parallel (`torchrun`, multi-GPU DDP), Google Cloud TPU v5e (8-device JAX), Mixed Precision (AMP), Weights & Biases (250 tracked runs), Git, Docker, Linux/Ubuntu.

Models Built from Scratch: Autoregressive Decoder Transformer (GPT-1), Rectified Flow / Flow Matching with ODE solvers, Conditional DDPM, WGAN-GP (gradient penalty & critic loop), DCGAN, Mamba-Attention hybrid (DSMaT).

AWARDS, CERTIFICATIONS & ACHIEVEMENTS

- **Undergraduate Thesis Distinction:** Awarded **Grade A** (4.00/4.00) in thesis defense, Department of CSE, IIUC (2026).
- **Academic Recovery & Final-Year Honors:** Attained **GPA 3.702 / 4.00** in final semester, overcoming early-year academic obstacles.
- **DeepLearning.AI Specializations (Coursera):** Neural Networks & Deep Learning, Improving Deep Neural Networks (Tuning, Regularization, Optimization), Convolutional Neural Networks, Sequence Models.
- **Resourcefulness in Constrained Research:** Conducted ~81 GPU-hours of research experiments across 250 logged runs (including multi-GPU DDP and an 8-device TPU/JAX port) solely on free-tier compute infrastructure without access to a local GPU.

ACADEMIC REFERENCES

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